

What is Data Analytics?

Data Analytics is the process of collecting, cleaning, transforming, analyzing, & interpreting data to find useful information & support decision-making.

Step 1 → Data Sourcing

- collecting the required data.

- Primary data - data collected directly.

- secondary data - data already collected by someone else.

- sources: Surveys, feedback, existing datasets, etc.

Step 2:- Data Preprocessing / Data Transformation

- cleaning and preparing the data before analysis.

- Handling missing or ~~into~~ incorrect data.

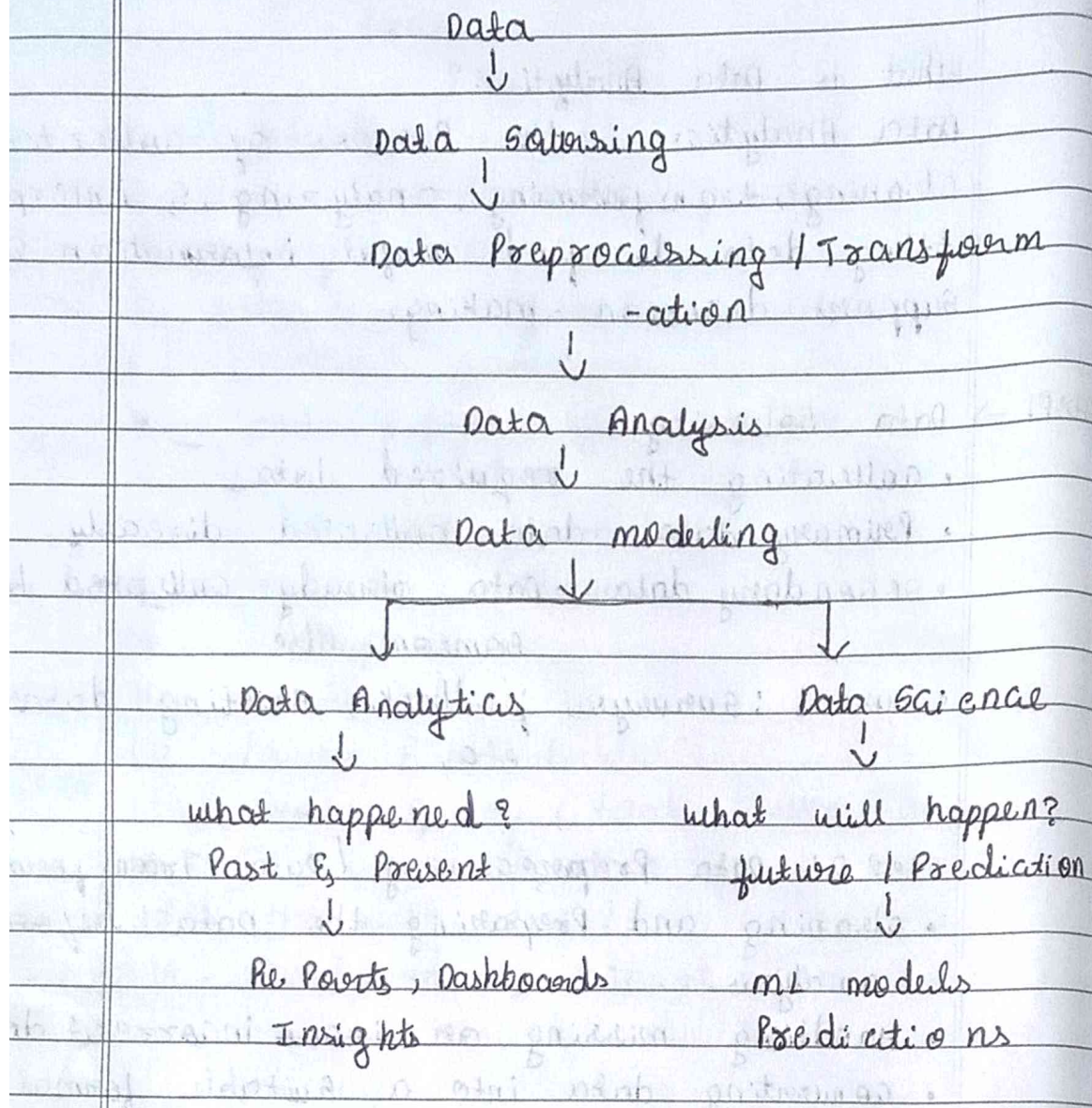
- converting data into a suitable format.

Step 3:- Data Analysis

Examining the data to find patterns, trends and useful information.

Data modeling

Data modeling involves building relationships between tables and organizing data properly for analysis.



→ Tools and Technologies

1. ~~Data~~ Excel
 2. MySQL
 3. Power BI
 4. Tableau
 5. Python
 6. Deep Learning
 7. Generative AI
 8. Machine Learning
- } → Data Storage & preprocessing
 } → Data Presentation & visualization

Introduction to Excel

→ Definition

Microsoft Excel is a powerful spreadsheet + tool used to store, organize, analyse & visualize data. In data analytics it acts as a beginner friendly yet powerful for handling real world datasets.

→ Why is Excel important?

- easy to learn and use.
- handles structured data efficiently.
- Provides many formulas and functions.
- supports charts and data visualization.
- can perform data analysis without programming.

→ Key components of Excel

1. workbook and worksheets

work book → An Excel file is called a workbook

work sheet → A Page inside a workbook is called a worksheet.

2. Rows and Columns

Rows:- Represented by numbers (1, 2, 3, 4 --)

columns:- Represented by letters (A, B, C, D --)

3. Cell

The intersection of a row and a column is called a cell.

example :- B3 $\rightarrow$ Column B + Row 3

$\rightarrow$ Types of data in Excel

1. Text data :- Text contains letters, words or a combination of letters & numbers

used for :- names, categories, address, etc.

2. Numbers :- numbers are values that can be used for calculations

used for :- salary, marks, quantity, etc..

3. Dates & Time :- Excel stores dates & time in a format that can be used for calculations.

eg :- 02/09/2026

• 10:30 Am

used for :- Birth dates, sales dates, etc..

4. Boolean data :- Boolean data has only two possible values : True or false

used for :- yes/no conditions, etc..

5. Geographical data :- represents location
used for :- maps, location analysis, etc..

-> Important Excel features for DA

1. Data Extraction

used to collect or retrieve required data from different sources.

examples :-

- import data from CSV files
- copy data from other worksheets

2. Data cleaning / Data Transformation

used to remove errors and make data ready for analysis.

common Tasks:

- Remove duplicates
- Handling missing values
- correct spelling
- Removing the irrelevant data & adding necessary data
- Text columns
- find and replace

3. Formulas and Functions

used to calculate and analyze data.

- Basic -> sum, average, count
- logical -> if, and, or, not
- Lookup -> Lookup, xlookup, vlookup, hlookup.
- Text -> concat, upper, lower

4. Data visualization

used to represent data graphically so that trends and patterns are easier to understand.

- charts :- Bar, Column chart, Line, Pie, etc.

- conditional formatting
- Spikes

4. Pivot Tables

A Pivot Table is used to summarize & analyze large amounts of data quickly. it can show:- Total, count, Average, minimum, maximum,

eg:- Finding Total Sales for each product or each month

5. Sorting and Filtering

Sorting:- Arranges data in a specific order

eg:- • A $\rightarrow$ Z , Z $\rightarrow$ A

• Smallest $\rightarrow$ Largest

Filtering:- Shows only the data that matches a condition

eg:- show only:
Department = IT

6. Data validations

used to control what type of data can be entered into cell.

eg:- • create a dropdown list.

• Allow only numbers.

• Restrict invalid entries.

• Allow dates within a specific range.

→ Advantages of Excel

- Beginner Friendly
- Fast For small to medium datasets.
- widely used in companies.
- integrates with tools like PowerBI.

→ Disadvantages of Excel

- not suitable for very large datasets
- limited automation compared to coding

Functions

Set of Instructions / set of code is already predefined is called function

→ There are 3 main components of function

1. Function name
2. number of Input
3. return value

1. Function name:- The Function name tells Excel what calculations or operations you want to perform.

ex:- • sum → adds numbers

• MAX → finds the largest value

• MIN → finds the smallest value

ex:- =sum (A1:A10)

sum is the function name.

2. Number of Input:- Inputs are the values or cell ~~refer~~ references that we give to the function so it can perform the calculation.

eg:-

=sum (A1:A10)

here, A1:A10 is the input given to the sum function.

Different functions can accept different number of inputs

3. Return value

The return value is the result that Excel gives back after processing the inputs.

eg :-

= sum (10, 20, 30)

- function name : sum
- Inputs : 10, 20, 30
- Return value : 60

→ Excel Text functions

1. LOWER()

Defination :

converts all letters in a text into lower case.

Syntax :-

= LOWER (text)

eg :- = Lower (LAKSHMI)

o/p :- lakshmi

2. UPPER()

Defination :

converts all letters in a text into uppercase

Syntax :-

= UPPER (text)

eg :-

= UPPER (lakshmi)

o/p :- LAKSHMI

3. PROPERC)

Definition:

converts the first letter of each word into uppercase and the remaining letters into lowercase

Syntax :

= PROPER(text)

eg :

= PROPER(Heamanth kumar)

O/P :- Heamanth kumar

4. number of characters - LEN()

counts the total numbers of characters in a text, including spaces.

Syntax:

= LEN(Text)

eg :

= LEN(Lakshmi)

O/P :- 7

5. LEFTC)

Extracts a specified numbers of characters from the left side of a text.

Syntax :

= LEFT (text, num. chars)

A1 = Lakshmi

eg :- = LEFT (A1, 3)

O/P :- Lak

6. RIGHTC)

Extracts a specified number of characters from the right side of a text.

Syntax :

= RIGHT (text, num-chars)

eg :- A1 = Lakshmi

= RIGHT (A1, 3)

o/p = hmi

7. Extract 2 characters from 5th position

Extracts a specific number of characters starting from a given position in the text.

Syntax :

= MID (text, start-num, num-chars)

eg :- A8 = Hemanth

= MID (A8, 5, 2)

o/p = NT

8. Separate name into First half & Second half

Suppose A1 contains : LAKSHMI

There are 7 characters, so we can split it into approximately two halves.

First half

Definition :- Extracts characters from the beginning of the text up to the middle.

Formula:

$$=LEFT(A1, LEN(ROUNDUP(LEN(A1)/2, 0)))$$

Result = Laks

second half

Extracts the remaining characters from the middle to the end.

formula:

$$=RIGHT(A1, ROUNDDOWN(LEN(A1)/2, 0))$$

Result: hmi